Lab 11b: More Op Amp Applications: Active Filter; PID Motor Control

REV 3: fix I-source references; revise ckt to adjust overall gain rather than P separately: add annotating "balloons" to figures, & explan. of D gain; March 14, 2002

Re: active filters: Chapter : 5.01 - 5.10: skim most of this, but read closely the sections that concern the active filter you will build: the passive and VCVS: 5.01, 5.03-5.06 and first pages of 5.07

1 Introduction

Today's lab invites you to look at two useful circuits that flirt with instability. In the first, the filter circuit, the stability issue is incidental; in the PID circuit (where the circuit response includes "proportional," "integral," and "derivative" of the circuit error) , stability is the central issue. The second circuit, the PID, will give you a chance to apply several subcircuits that you have met before (integrator, differentiator, summing circuit, push-pull brought within feedback loop), plus a new one (differential amplifier). This exercise provides a first chance to use multiple op amps in one larger circuit.

2 An Active Filter: VCVS (45 min.)

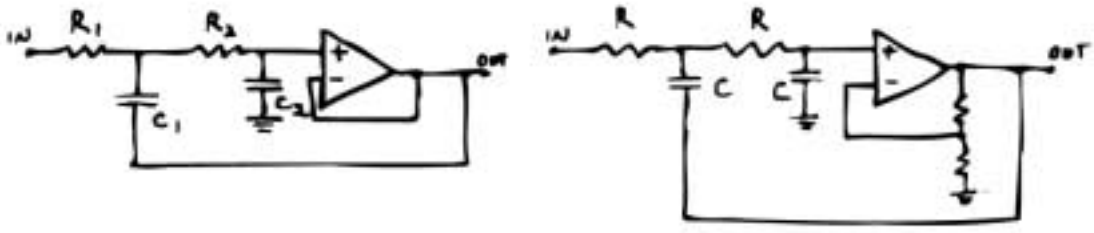


Figure 1: Two forms of a 2-pole active low-pass filter

Both circuits, above, work fundamentally the same way, feeding back a boost in a frequency band around f_{3dB}. We ask you to build the right-hand circuit, because this form–which the Text calls"VCVS"–is easy to build and then easy to tune. The VCVS form lets us use just one R value and one C, and then vary the positive feedback by adjusting the gain of the op amp circuit:

2.1 "Flattest" Response: versus passive RC

Wire the circuits below, with $R_{GAIN} = 4.7k$. If you have a *resistor substitution box* available, use it to form the 4.7k feedback resistor.

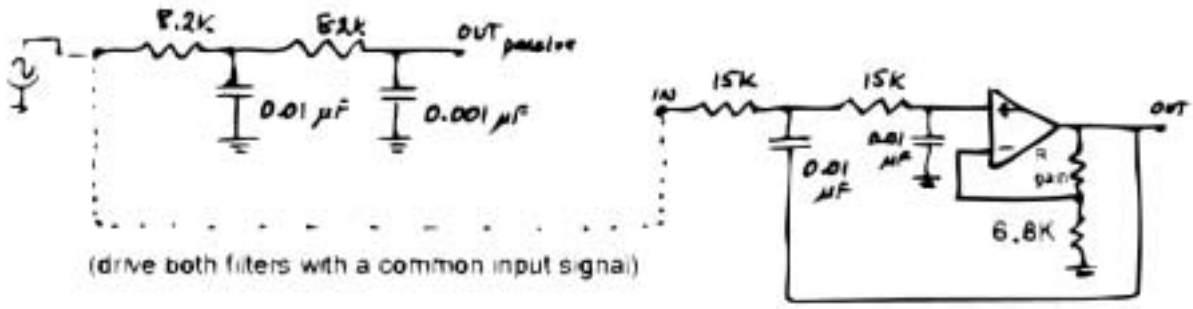


Figure 2: VCVS: a particular implementation, with cascaded RC's added for comparison

Confirm that the circuit behaves like a low-pass; note f_{3dB}, and note attenuation at 2 X f_{3dB} and 4 X f_{3dB}. We hope you will find the"-12dB/octave" slope that is characteristic of a"2-pole" filter, though you won't see that full steepness in the first octave above f_{3dB}. We hope, also, that the simple cascaded RC looks wishy-washy next to this improved filter. The simple RC and the active filter should show the same f_{3dB}.
Presumably you have been watching the outputs of both filters on the scope, as you drive the two filters with a common input. For a vivid display of the two filters' frequency-responses you will want to *sweep* frequencies automatically. You have done this before, using the function-generator's *sweep* function, but this time you must do the task a little differently from the way you did it earlier. This time, you *cannot use the X-Y* display mode. Instead, use a conventional sweep (this allows you to watch two output signals, not just one), while *triggering* the scope on the function generator's RAMP output (use the steep falling edge of the ramp).

2.2 Effects of Varying the amount of Positive Feedback: other filter *Shapes*

Once you have a pretty display of a flat passband, try altering the filter shape: in place of the 4.7k feedback resistor (which helps define the op amp circuit's *gain*), try the following values (this change of gain is very easy if you are using a resistor substitution box). The 4.7k is shown again, in the table below, to make the point that 4.7k provides an intermediate behavior.

Filter Type	**R2**	**Gain**
best time delay (Bessel)	1.8k	1.3
flattest (Butterworth)	4.7k	1.6
steep, 2dB ripple (Chebyshev)	7.5k	2.1
nasty peak (no one claims this one!)	12K	almost 3
OSCILLATOR!	15K	>3

The last case, in which we deliberately overdo the positive feedback, is pointless in a filter–but for today's lab it may be useful: it reminds us of the boundary we are moving toward in this lab, where positive feedback becomes harmful (a page or so farther, just below).

2.3 Step Response; waveform distortion

(Text sec. 5.05, esp. pp.271-272, re*: Bessel filter)*
Watch the circuit's response to a 200Hz square wave, and note particularly the *overshoot* that grows with circuit gain. If you are feeling energetic, you might test also the claim that the tamest

of the filters (with R = 1.8k), which shows the best step response, also shows the least waveform distortion. The R=7.5k filter should show most distortion. Try a triangle as test waveform. The contrast will not be very striking: we saw only a little distortion, from the worst of the filters.

3 PID Motor Control

The task we undertake here looks simpler than it is. All we aim to do is control the position of a DC motor's shaft, by letting it drive a potentiometer and feeding back the pot's voltage. Here's the scheme:

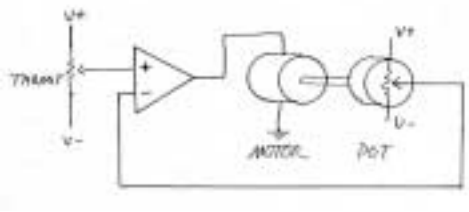


Figure 3: Basic Motor-Position Control Loop: Very Simple!

What could be simpler? Not much, on paper. But the challenge turns out to lie in keeping the circuit *stable*. The issue is fundamentally the same as the one you met in Lab 10b, when you noticed that a low-pass in an op-amp's feedback loop could turn negative feedback into *positive,* if we weren't careful. The problem arose from the fact that an op amp provides -90 degrees of phase shift, so that just 90 degrees more can get us into trouble. Another way to say that—and a way that may be more appropriate to today's circuit—is to note that the op amp acts like an *integrator*, above a few *tens*of Hz. This *integration* effected by the naked op amp results from the internal "compensation" that rolls off its gain so as to keep the feedback circuit stable.

In today's circuit we are stuck with a similar -90 degree shift, or an integration. This time, it comes not from the op amp. We avoid that effect by not using the "naked" op amp, and so can hide from its phase shift. Instead, the integration comes from the nature of the stuff we are putting inside the loop: a motor whose shaft position we are sensing. We drive the motor with a voltage, which controls its speed; it spins for a while; the position it achieves is the time integral of the spin rate. That last proposition means we are *stuck with an integration inside the loop.*

To make this last point graphically vivid, here's a scope image showing how the position-pot responds to a *square-wave* input to the motor. The triangular output looks a lot like what you saw in Lab 9's integrator, doesn't it? —apart from the inversion that Lab 9's op-amp integrator inserted.

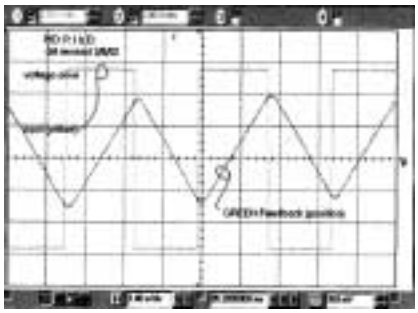


Figure 4: Motor-drive to Position-sensing Potentiometer forms an INTEGRATOR

Here's a block diagram of our feedback scheme, with the variables indicated, as they work their way around the loop. We'll show it first in the more formal control-loop form used in our class discussion:

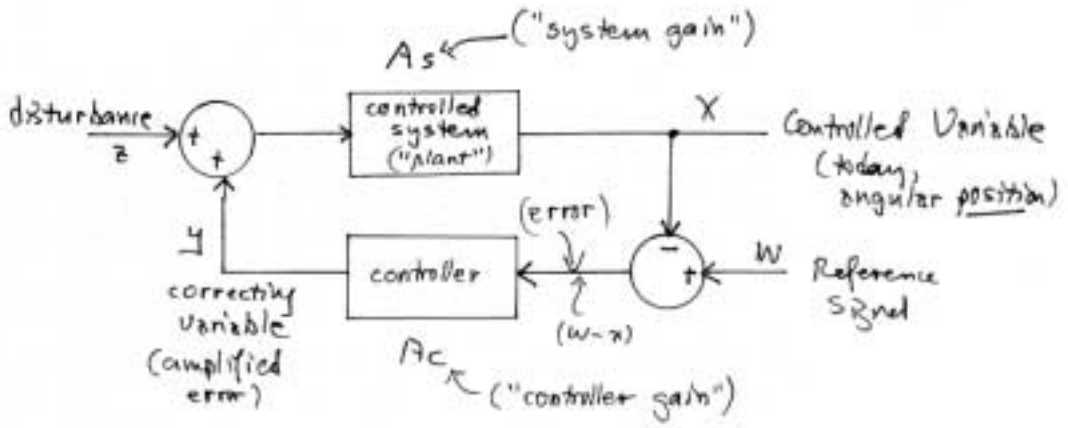


Figure 5: Loop to control motor shaft position: block diagram in form used in class discussion

Now, here's the same diagram redrawn to look more like the op amp loops that we are accustomed to seeing:

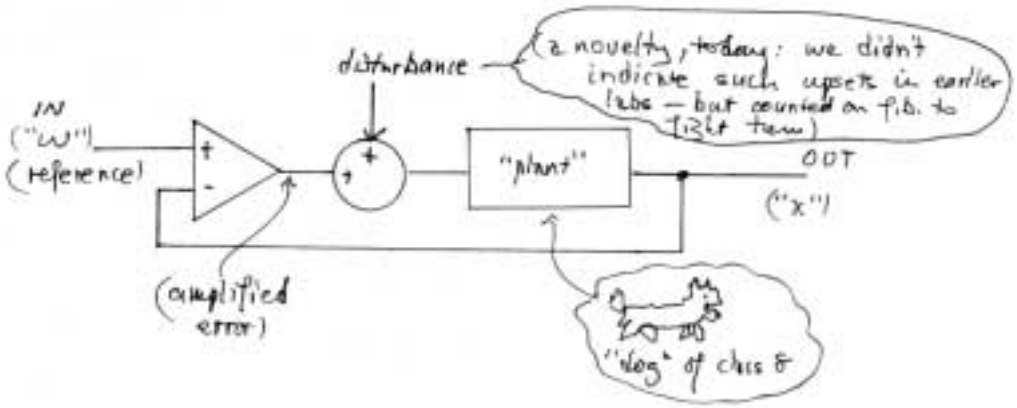


Figure 6: Loop to control motor shaft position: block diagram in form familiar from op amp discussions

Stability We will find that we can make the circuit at least marginally stable, simply by keeping the *gain* of the circuit low enough: more precisely, we'll keep the *loop gain, AB* low enough. In Lab 10b, we saw the same pattern—in which stability improved when *loop-gain was reduced*; in those Lab 10b experiments, we could not control the gain of the op amp ("A"), so we varied the fraction fed back ("B").

Just below is the circuit where we met the effect: capacitive load could make the circuit unstable, but cutting gain ("B") could restore stability.

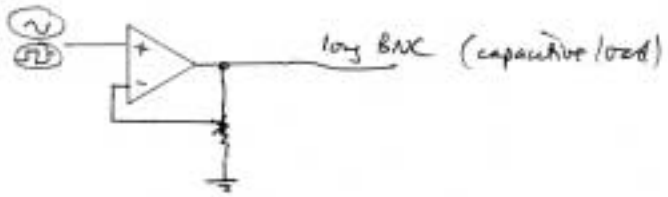


Figure 7: Lab 10b circuit: diminishing signal fed back was able to stabilize circuit despite C-load

The signal fed back was shrinking as its phase shift was growing more dangerous (approaching the -90 degrees that could bring on oscillation). So, when we attenuated it further (with a voltage divider, in exercise 10b-2), we were able to keep our circuit stable.

Today, we again regulate loop gain (AB), but we will do this, first, by varying "A," the "amplified error" term, rather than "B". (We will be able to vary "A" because we replace the usual very-high-gain op amp with a *pseudo-op amp* made from a differential amp followed by a gain block). We will not play with "B," the fraction fed back.

Once this "P"-only loop is wired, we will try gradually increasing the gain —much as in the active-filter case. We should find the circuit fairly stable for low gains, then as we increase gain we should begin to see overshoot and ringing, evidence of the circuit's restlessness; at still higher gains, the circuit should oscillate continuously.

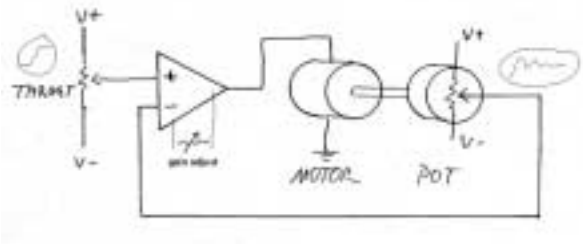


Figure 8: Proportional-only drive will cause some overshoot; gain will affect this

At the end of these notes we attach some scope images describing just such responses to variations in simple "proportional" gain.

3.1 Motor Driver

Let's start with a subcircuit that is familiar: a high-current driver, capable of driving a substantial current (up to a couple of hundred milliamps). We'll use the power transistors you've met before: 2N3055 (npn) and 2N2955 (pnp). The motor presents the kind of troublesome load likely to induce parasitic oscillations, as in the last exercise of Lab 10b. We need, therefore, the protections that we invoked there: not only decoupling of supplies, but also both a *snubber* and *high-frequency feedback* that bypasses the troublesome phase-shifting elements.

We are trying hard, here, to *decouple* one part of the circuit from the others: the 15μF caps should prevent supply disturbances from upsetting the *target* signal. Similar caps at the ends of the motor-driven potentiometer aim to stabilize the feedback signal. We also suggest that you use an *external power supply* to provide the *motor's* $\pm$15V supplies; we do this not for decoupling, but because the motor's maximum current exceeds the breadboard's 100mA rated output, and might have disturbed those supplies even if we decouple fanatically. The external supply can provide the necessary current.

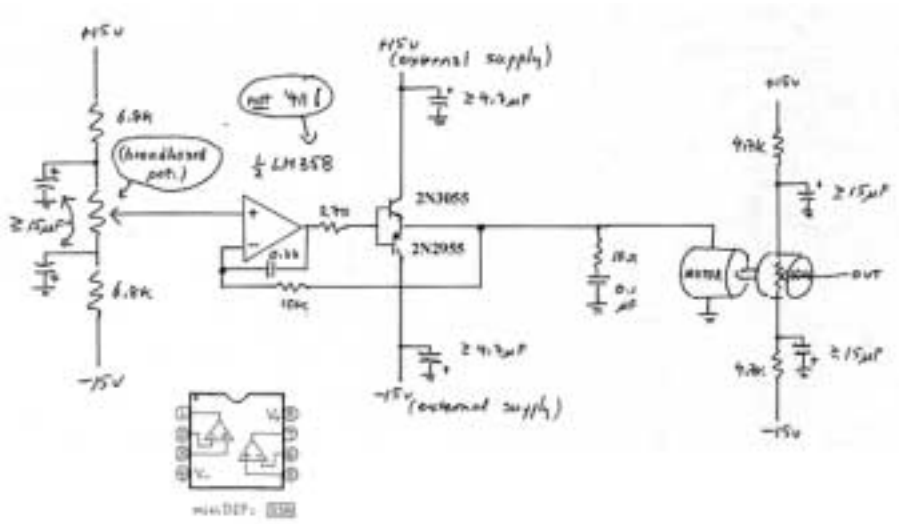


Figure 9: Motor-driver

Wire up the two potentiometers, as well. The resistors at the ends of the two potentiometers—6.8k resistors on input, 4.7k resistors on the motor pot—restrict input and output range to a range of about ± 10V, so as to keep all signals within a range that keeps the op amps happy. The difference in R values makes sure that the input range cannot exceed the achievable output range.

You can test this motor driver by varying the input voltage, and watching the voltage out of the motor-driven pot. Any V_{IN} more than a few tenths of a volt should evoke a change of output voltage. You will hear the motor whirring, and will see the shaft slowly turning (the motor drive is geared down through a two-stage worm- and conventional- gearing scheme). A clever clutch scheme allows the motor to slip harmlessly, when the pot reaches either end of its range. If the *signs* of V_{IN} and the *change* in V_{OUT} *do not match*, then be sure to interchange leads of one of the pots, so as to make them match. We don't want a hidden inversion, here, to upset our scheme when we later close the loop.

3.2 Pseudo Op Amp

Now we do a strange thing: we use a pair of op amps to make a rather-crummy op-amp like circuit. Here is our "pseudo op amp."

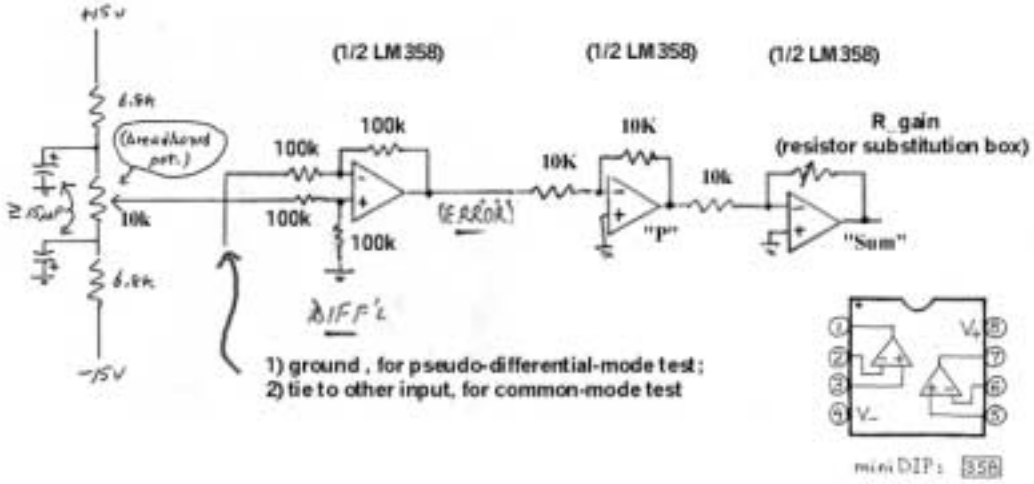


Figure 10: Differential Amp Followed by Inversion and a Gain Stage

The first stage you recognize as a standard differential amp. It shows unity gain. The second stage simply inverts[1]; the third stage seems to be doing no more than *undoing* the inversion of the preceding circuit. That is true, at this stage; but we include this circuit because soon we will use it, fed by two more inputs, as a *summing* circuit. So used, it will put together the three elements of the *PID*controller: Proportional, Integral, and Derivative.

This circuit is a differential amplifier, with gain that is adjustable, but never anything like so high as what we are accustomed to in op amps. We need this modest gain, and we need a virtue of this simple circuit: no appreciable phase-shift between input and output. Both characteristics contrast with those of an ordinary op amp, as you know: the ordinary op amp shows fixed, high gain, and *integrator* behavior beginning at 10 or 20 Hz. We cannot afford to include such an integrator in our loop, because—as we have noted above—we are stuck with another integration, and two integrations in series would get us into trouble, turning negative feedback into positive.

We suggest that you use a resistor substitution box to set the *summing* circuit's gain. Set the gain at 100 ($R_{FEEDBACK}$ in summing circuit = 1M), and see whether a *common-mode* signal—a volt or so applied from the input pot, applied *common-mode* to both inputs simultaneously—evokes the output you would expect. (Do you expect *zero* output?) Then ground one input, so as to apply a "pseudo-differential" signal, and see if you get the expected gain of -10, in response to a change at the input pot.

A DVM may be handier than a scope, at this point, to confirm that the output of this chain of three op-amp circuits shows a pseudo-differential gain of +10, while you drive the input with the *input* potentiometer voltage. When you finish this test, leave the output voltage close to zero volts.

[1]This inversion is included so as to let this signal share a polarity with the "Derivative" and "Integral" signals, soon to be generated; these signals will come from circuits that necessarily invert.

3.3 Drive the Motor

You have already tested the motor driver. Let's now check the three new stages—the pseudo op amp—by letting its output feed the motor-driver. Confirm that you can make the motor spin one way, then the other, by adjusting the input pot slightly above and then below zero volts. (The motor-driven pot fortunately can take the pot to its limit without damaging pot or motor: the motor continues to spin, once the pot has hits its stop; the motor's gearing has been cleverly designed so as to permit this slippage.)

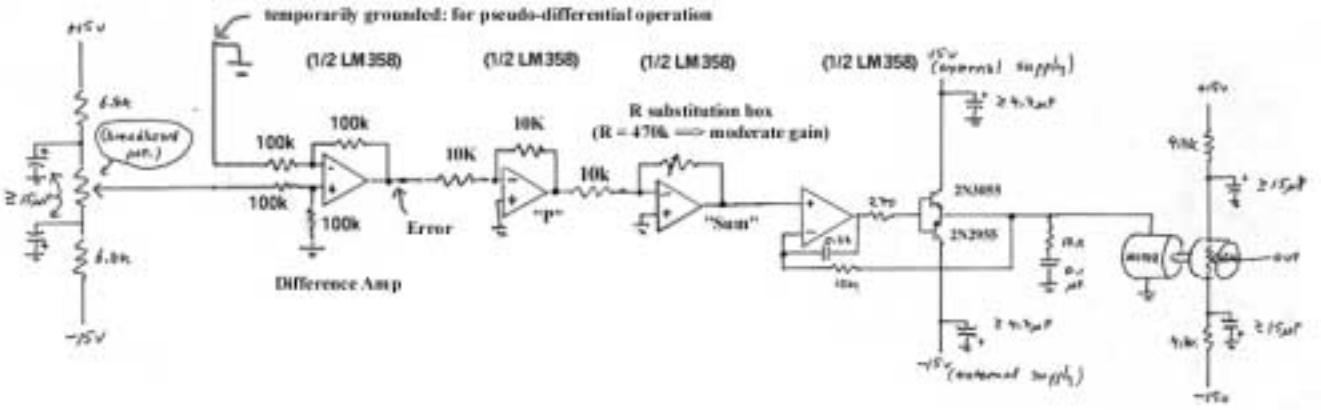


Figure 11: Try making motor spin, to test the diff amp, gain stage, sum and motor drive

3.4 Close the Loop

Now reduce the *gain*, using the *R substitution box*: set gain to about 50 (R_{Sum} = 470k). Replace the *ground* connection to the inverting input of our "pseudo op amp" with *the voltage from the output potentiometer*.

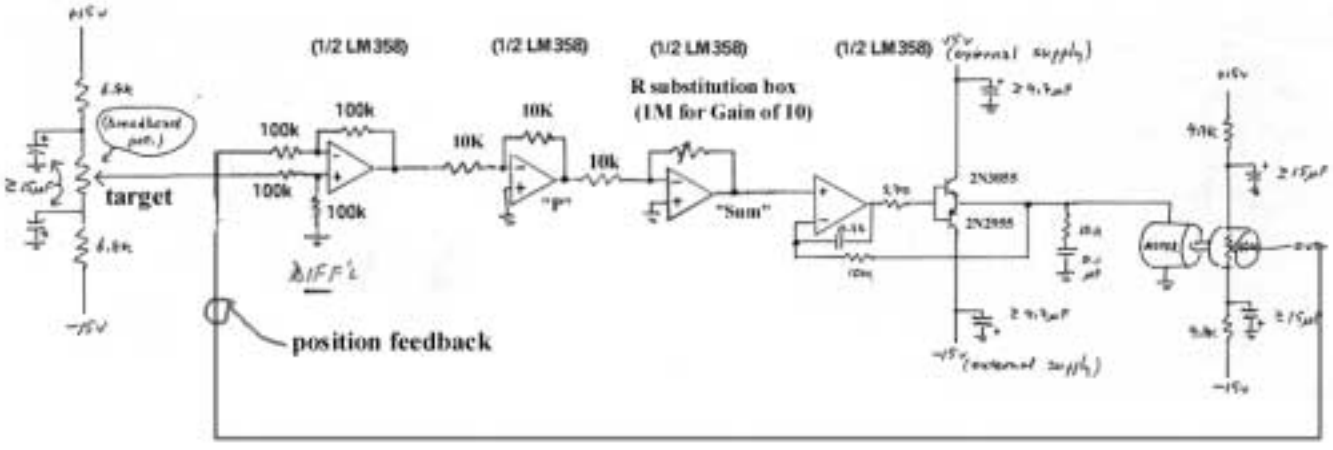


Figure 12: The loop closed, at last: Proportional only

Watch V_{in} on one channel of the scope, $V_{output-pot}$ on the other channel. *If a digital scope is available, this is a good time to use it*, because a very-slow sweep rate is desirable: as low as 0.5 second- or even 1 second -per division.

Manual or Function-Generator Steps? A function generator, providing a small square wave (±0.5V, say), at the lowest available frequency (about 0.2Hz) can provide your test input. Alternatively, you can manually apply a "step input" from the *input* pot: a step of perhaps a volt. The *output* pot should follow—though showing a few cycles of overshoot and damped oscillation. A second way to test the circuit's response is to leave the input constant, then force the pot away from its resting position, simply by turning the knob of the *output* pot. Let go, and the knob should return to its initial position—but showing some overshoot and oscillation, as when the change was applied at the input pot.

Start with a very low gain, which should make the circuit stable, even in this *P-only* form. Try $R_{Sum} = 100k$; now use the substitution box to dial up increasing gain. At $R_{Sum} = 220k$ we saw some overshoot and a cycle or two of oscillation. That oscillation is evident in the motion of the motor and pot shaft; if this shaft were controlling, say, the rudder of an airplane, this effect would be pretty unsettling. The circuit works—but it would be nice if we could get it to settle faster and to overshoot less.

Increasing the gain, at $R_{Sum} = 680k$, we were able to make out several cycles of oscillation (the bigger, uglier trace shows the motor drive voltage; there the oscillation is more obvious):

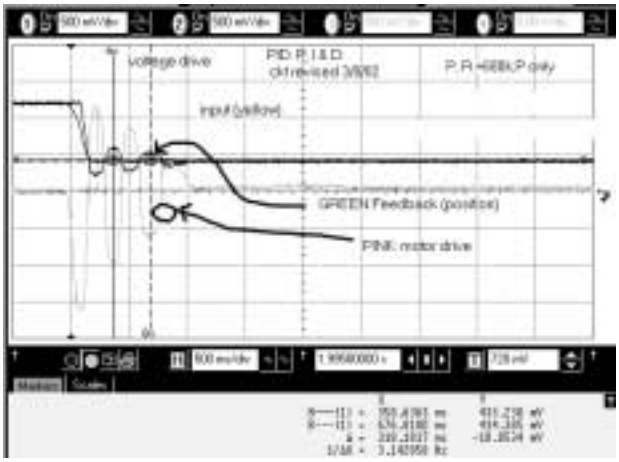


Figure 13: P only: gain is high enough to take us to the edge of oscillation

With a little more gain ($R_{Sum} = 1M$, in our case) and the application of either a step change at the input, or a displacement of the output pot by hand we saw a continuous oscillation. Find the gain that sets your circuit oscillating, and then *note the period of oscillation*, at the lowest gain that will give sustained oscillation. We will call this the period of "natural oscillation," and soon we will use it to scale the remedies that we'll apply against oscillation.

4 Adding a Derivative of the Error

Well, of course we *can* get it to settle faster; we *can* improve performance. (If we couldn't, would the name of this sort of controller include the "I" and "D" in its name, 'PID"?)

We can speed up the settling markedly, and even crank up the "P" gain ("proportional") a good deal once we have added this derivative. Thinking of the stability problem, as we did for op amps generally, as a problem of taming phase shifts of sinusoids, we can see that inserting a derivative into the feedback loop will tend to *undo an integration*, at least in some degree.

The integrations are the hazard, here: one is built in, the translation from motor rotation to motor

position. Additional integrations lagging phase shifts can carry us to the deadly minus-180-degree shift that turns nice feedback into nasty, and brings on the oscillation you have just seen.

4.1 Derivative Circuit

The standard op amp differentiator shown below can contribute its output to the summing circuit. Here, we show the entire prior circuit, with the differentiator added. Its gain is rolled off at about 160Hz.

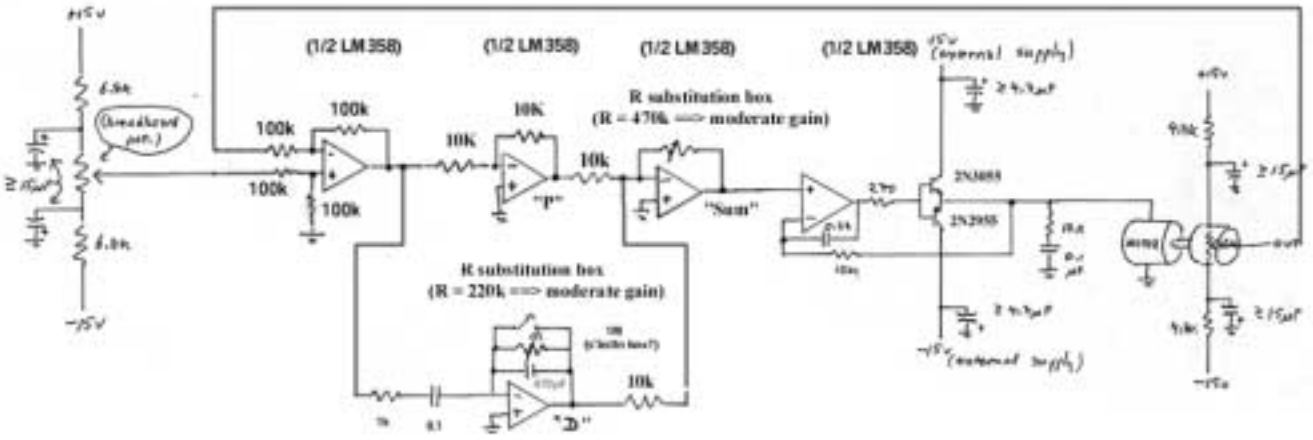


Figure 14: Derivative added to Loop

How Much Derivative?

Our goal, in adding derivative, is to cancel the extra phase shift otherwise caused by a low-pass effect that brings on instability. How do we know at what frequency this trouble occurs? We got that information by looking at the frequency (or period) of oscillation, back in section 3.4, when you gradually increased the *P-only* gain till you got either long-settling disturbances or a continuous oscillation, in response to a disturbance. When we did this, we got a period of roughly 0.3 second.

This instability results from the phase shift caused by a *second* low-pass or integration, in addition to the implicit integration that occurs because of our feeding back pot *position*, an integration of pot *rotation rate*, the characteristic that our driver controls. We want to inject enough derivative so as to *undo* or cancel this second low-pass effect, thus avoiding the dangerous phase shift that would result.

Our goal is to arrange things so that the derivative contribution, D, is equal to the Pcontribution, at the frequency where trouble otherwise would occur. The D should keep the loop stable, until yet another low-pass cuts in; at this cut-in frequency, we should have made sure the loop gain is too low to permit an oscillation (less than unity).

Let's start by reminding ourselves what we mean by differentiator "gain;" then we'll calculate what RC we need for stability. Gain for a differentiator, by definition is

$$V_{OUT}/(dV_{IN}/dt)$$

. We know that

$$V_{OUT} = I \times R_{Feedback}$$

and this I is just .

$$C \times dV_{IN}/dt.$$

So

$$V_{OUT}/(dV_{IN}/dt) = \frac{R \times C \times dV_{IN}/dt}{dV_{IN}/dt} = RC.$$

So, RC defines the differentiator's *gain*. A differentiator's output amplitude grows linearly with frequency; V_{OUT}, in other words, for a sinusoid is proportional to ω:

$$V_{OUT_{Deriv}} = \omega RC$$

We want this $V_{OUT_{Deriv}}$ to equal $V_{OUT_{Proportional}}$—which is just equal to V_{IN} ("V_{IN}", to both "P" and D" circuits, is the *Error* signal: the output of the diff amp).
If we set these quantities equal, then

$$\omega RC = 1$$

$$2\pi f RC = 1$$

$$RC = 1/(2\pi f).$$

In words, this suggests that RC should be about 1/6 of the period of natural oscillation. In our case, where $T_{Oscillation} = 0.3s$, we'd set RC to about 0.05 s, or a bit less.[2] If we use a convenient C value of 0.1 μF, the our observations ($f_{natural-osc} \approx 3Hz$) seem to call for R of about 0.5M.[3] Let's make this value adjustable, though—because we want to be able to try the effect of more or less than the usual derivative weight: if you have a second *resistor substitution box*, use it to set the differentiator's gain (RC). Otherwise, use a 1M variable resistor. Watching the position of the rotator will let you estimate R to perhaps 20 percent; the midpoint value certainly is 500k, and 700k is close to the 3/4-rotation position. The differentiator's output goes into the summing circuit installed earlier, through a resistor chosen to give this "D" term weight equal to the "P"'s. We hope you will find this "D" to be strong and effective medicine. Once it has tamed your circuit's response—eliminating the overshoot and ringing—crank up the "P" gain, taking it to the gain that just set off oscillation in the "P-only" circuit. Is the circuit stable? If not, try more "D." Does an excess of "D" cause trouble? Incidentally, you can also use the scope image's *time-domain* image of the circuit's response to judge whether you have too much or too little D: too little, and you'll see remnants of the overshoot you saw with "P-only"; too much D, and you'll see an RC-ish curve in the output voltage as it approaches the target: it chickens out as it gets close.

Switch The toggle switch across the feedback resistor will let us cut "D" in and out; the switch seems preferable to relying, say, on a very-large variable R to feed the summing circuit. We find it hard to keep track of multiple pot settings, to know whether we're contributing "D" or not. A switch makes the ON/OFF condition easier to note.

[2] See., e.g., Tietsche and Schenk (sp?)..... A less formal approach appears in St.Clair..... From his website, ..., one can download an interesting simulator that allows one to try his rules. See also, [Exeter, UK simulator]

[3] We must confess that our circuit worked better with about *half* this "D" gain; we hope you'll come closer to the target.

4.2 Add Integral

Adding the third term, the "I" of PID can drive residual error (a difference between the input pot voltage and the output pot voltage) to zero. In today's circuit, that residual error is hard to see on the scope, so adding "I" will not reward you as adding "D" did. Your best hope will come if you cut the "P" gain very low: try $R_{Sum} = 100k$, so that the circuit feedback ought to tolerate a residual error, when not fed an "I" of the error. If you have been using a function generator to provide step inputs to your circuit, now replace that signal source with the manually-adjusted pot input. Slow the scope sweep rate, to a rate that permits you to see the multi-second effect of the integration.

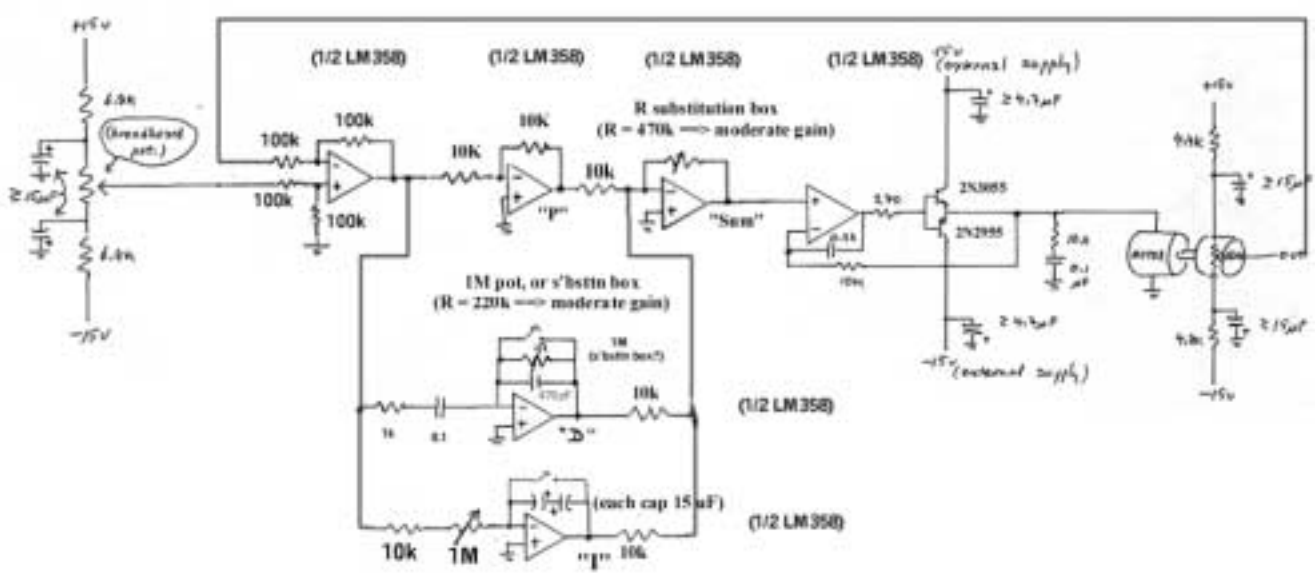


Figure 15: Integral added, to complete the PID loop

If you are using a digital scope, you will be able to watch input ("Target"), output ("Motor pot")and *Integrator* signals, after a step input applied from your input potentiometer. If you are patient, you can even make out the effects of the motor and pot's "sticktion": the motor and pot do not move smoothly in response to a slowly-changing input (the I term). Instead, it fails to move till I reaches some minimal level; then output voltage jumps to a new level, and waits for another shove. You can see these effects in some of the scope images attached at the end of these lab notes.

...but Stability may suffer It sounds dangerous, doesn't it?—tacking an integral term when integration, plus other lagging phase shifts, are just what threatens the circuit's stability. It *is* dangerous, as you can confirm by overdoing the "I". You should be able to evoke continuous oscillation, as in the dark days before you knew about the stabilizing effect of "D"!

Scope Images: Effect of Increasing Gain, in P-only loop

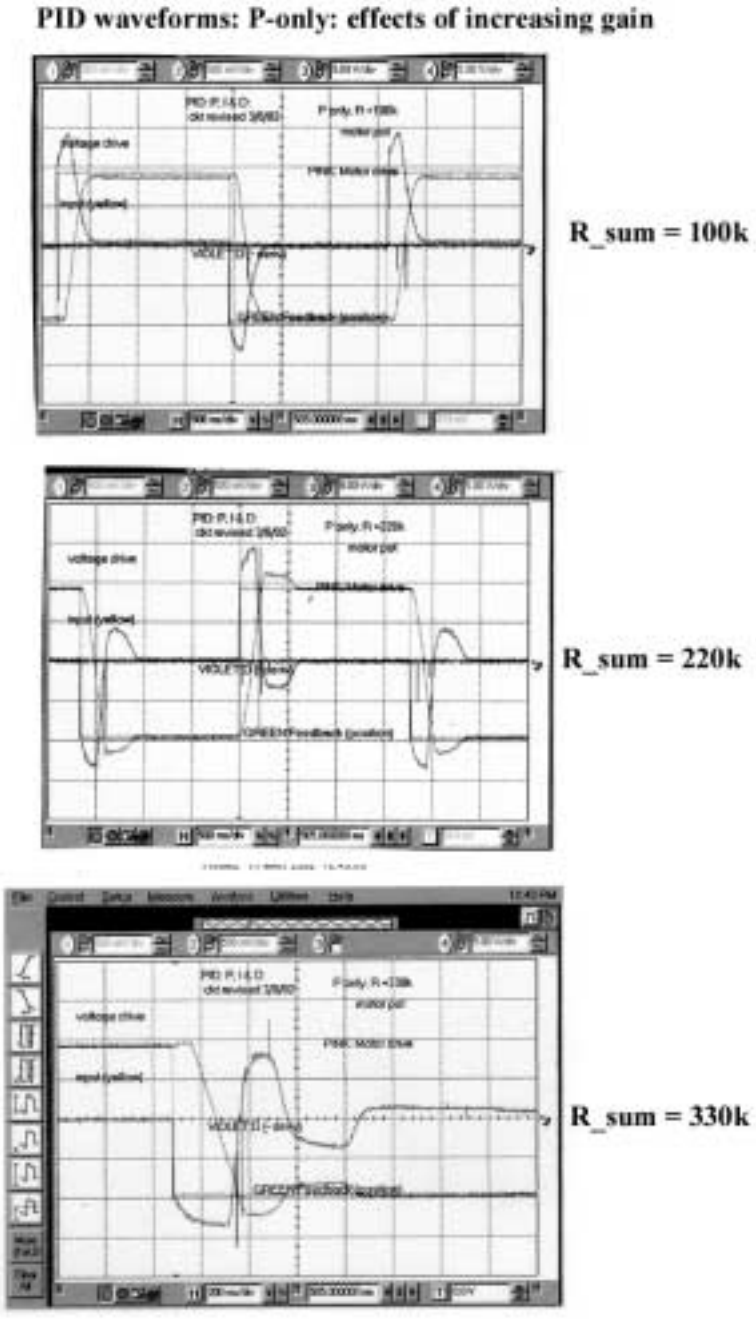
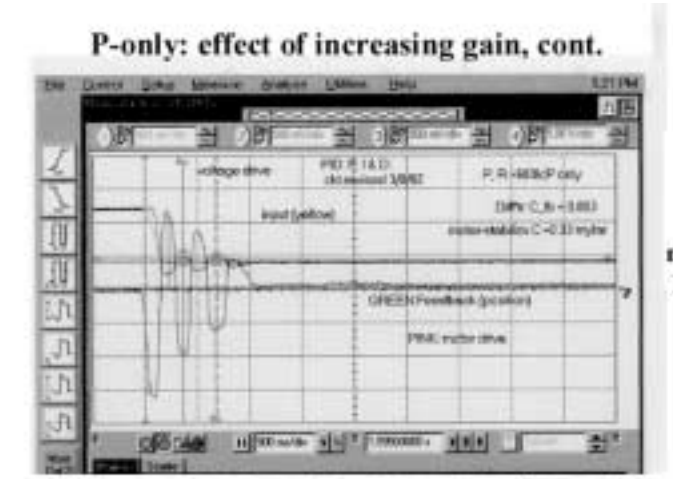
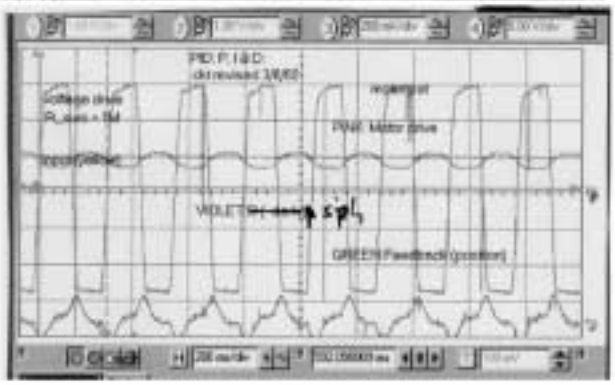


Figure 16: Increasing P-only gain brings increasing overshoot

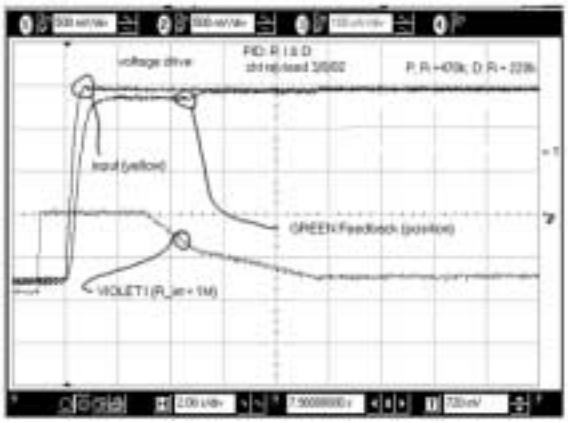


marginally stable:
R_sum = 680k



Unstable:
R_sum = 1M

Effect of integration term



Effects of "sticktion" visible

Figure 17: Increasing P-only gain, taken to brink of oscillation; and effect of integration term